HiFiAvatar: Prior-Guided Gaussian Appearance Learning for High-Fidelity Animatable Head Avatars from Monocular Video

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***Abstract*—Learning animatable head avatars from a short monocular video is attractive for expression-driven and pose-driven rendering, yet recovering fine subject-specific appearance from such input remains difficult. Existing Gaussian head avatar methods are efficient and practical, but many still learn point-wise appearance with weak constraints under monocular ambiguity, which often leads to over-smoothed results. This work presents HiFiAvatar, a monocular head avatar framework that improves appearance reconstruction by introducing a frozen subject-specific visual prior. The method first aggregates pre-trained visual features extracted from the training video into a compact identity and texture dictionary. Each Gaussian primitive then queries this dictionary from canonical space, fuses the retrieved cue with a learnable point embedding, and predicts its color through a lightweight surface-space decoder before standard Gaussian rendering. Because the prior is fixed during optimization, it acts as a stable regularizer that narrows the appearance solution space and reduces over-smoothing. Experiments on four benchmark datasets show that the proposed method achieves the best average results on three datasets and the best structural similarity and perceptual quality on the fourth, while preserving the efficiency of Gaussian avatar rendering. These results demonstrate more faithful recovery of high-frequency facial appearance from monocular video.**

***Index Terms*—animatable head avatar, monocular video, 3D Gaussian splatting, visual prior, perceptual quality**

I. Introduction

Animatable head avatars aim to render a person's appearance while remaining controllable by facial expression and head pose. They are increasingly important for digital humans, AR/VR, immersive telepresence, and virtual content creation. While studio-grade capture systems can achieve excellent quality [1]–[3], they are costly and hard to deploy at scale; in contrast, a short monocular video is easy to acquire and naturally contains motion cues required for reenactment. Yet, recovering rich high-frequency appearance from monocular supervision remains challenging.

Most monocular pipelines follow a common recipe: estimate pose/expression with a parametric head model (e.g., FLAME [4]) using monocular fitting methods [5]–[7], and then optimize a person-specific renderer across frames. NeRF-based methods [8]–[16] can deliver strong realism but are often expensive due to volumetric ray marching. Recently, 3D Gaussian Splatting (3DGS) [17] has emerged as an efficient alternative by rasterizing anisotropic Gaussian primitives, enabling optimization-friendly and real-time monocular head avatars. As a result, 3DGS has become a strong backbone for expression/pose-driven head avatar reconstruction [18]–[27]. However, a noticeable fidelity gap persists: high-frequency regions are still frequently softened in current monocular 3DGS avatars.

We argue that this "detail gap" is fundamentally an *appearance-learning bottleneck*. In the monocular setting, high-frequency appearance is intrinsically under-constrained: multiple geometry/texture explanations can produce similar renderings under limited viewpoints, so reconstruction losses alone tend to favor a "safe average" that washes out localized details. At the same time, many effective 3DGS pipelines still model appearance as free per-Gaussian attributes, such as RGB values or low-order spherical harmonics, and optimize these appearance parameters directly with photometric and perceptual losses [18]–[27]. This combination makes direct point-wise appearance optimization weakly constrained: each Gaussian is asked to explain fine subject-specific appearance with little stable guidance beyond image reconstruction, making over-smoothing a recurring failure mode. From this perspective, the key missing piece is not necessarily a more complex renderer, but a stronger subject-specific prior that can regularize how per-Gaussian appearance is predicted.

This motivates our core insight. The training video already contains repeated evidence of the subject's appearance, but existing monocular 3DGS pipelines do not explicitly convert that evidence into a fixed prior that can constrain point-wise color learning. Pretrained vision encoders, however, provide a natural way to distill such evidence into robust identity- and texture-level representations [28]. We therefore treat pretrained tokens extracted from the training video as an *identity/texture prior dictionary*: a compact set of subject-specific appearance prototypes produced by a frozen encoder and kept fixed during avatar optimization. Crucially, this prior does not rely on extra annotations, additional capture hardware, or test-time information. Instead, it uses the inductive bias of a frozen pretrained encoder to convert the training video itself into a stable regularizer for appearance learning.

Based on this idea, we introduce HiFiAvatar, a framework that regularizes direct Gaussian RGB appearance learning with

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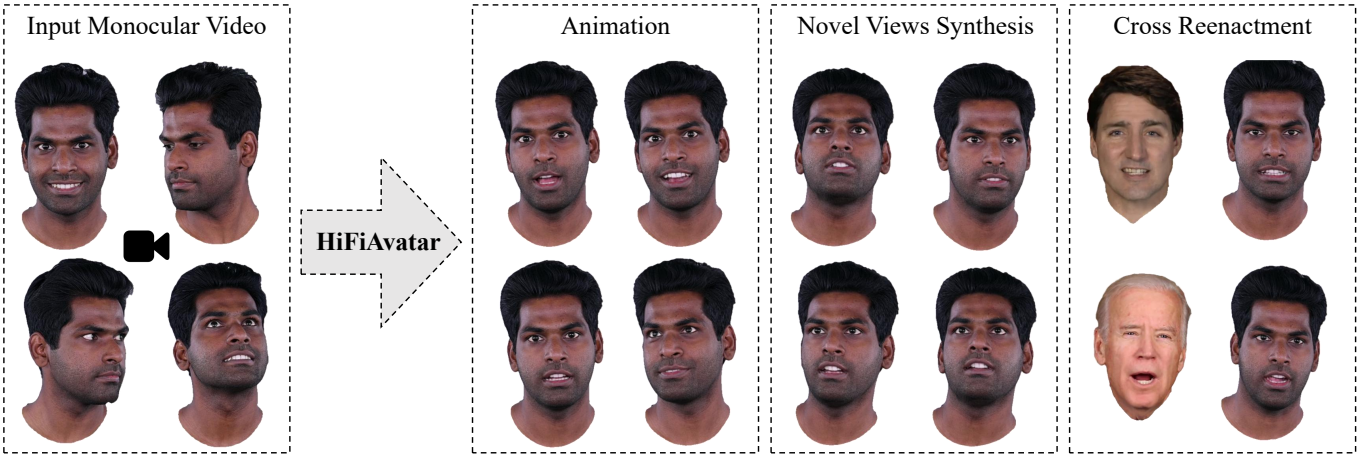

Fig. 1. From a short monocular video, HiFiAvatar reconstructs a high-fidelity animatable head avatar by combining a frozen identity/texture prior dictionary with prior-guided Gaussian RGB appearance learning, enabling more faithful recovery of fine appearance details.

a frozen identity/texture prior dictionary while keeping the standard 3DGS renderer unchanged. Each Gaussian queries the dictionary from canonical space, fuses the retrieved prior cue with a learnable base embedding, and predicts its RGB appearance through a lightweight surface-space decoder defined on the template UV layout. Unlike methods that treat Gaussian appearance mainly as free per-point parameters to be fit directly, our method still learns per-Gaussian RGB under reconstruction supervision but predicts it through a prior-guided appearance module constrained by a frozen subject-specific dictionary. This design injects subject-specific visual guidance exactly where the bottleneck arises, namely in point-wise appearance prediction before rasterization. As a result, it narrows the appearance solution space, makes the learned RGBs less prone to the safe-averaging behavior induced by monocular supervision, and does so without changing the image formation process of standard 3DGS. Figure 1 presents representative reconstruction results. Experiments on INSTA Dataset [29], PointAvatar Dataset [30], NerFace Dataset [9], and EmoTalk3D Dataset [31] show clear gains on INSTA, PointAvatar, and NerFace, and the best average SSIM/LPIPS on EmoTalk3D compared to strong 3DGS-based head avatar baselines.

Contributions. The contributions are summarized as follows:

- We introduce HiFiAvatar, a framework for high-fidelity animatable head avatars from monocular video that regularizes direct Gaussian RGB appearance learning with a frozen identity/texture prior dictionary.
- We build an identity/texture prior dictionary by aggregating pretrained visual tokens from the input video, query it from canonical Gaussian anchors, and decode the fused embeddings with local surface context to predict Gaussian RGB appearance.
- We validate the proposed prior-guided appearance learning strategy with strong baselines and ablations across four benchmark datasets [9], [29]–[31], demonstrating substantial gains in perceptual quality and high-frequency appearance reconstruction under monocular supervision.

II. Related Work

A. Monocular Animatable Head Avatars

Reconstructing an animatable head avatar from a short monocular video remains severely under-constrained because geometry, pose, expression, and appearance must be disentangled from limited viewpoints. Most monocular pipelines therefore rely on parametric head models, such as 3DMMs and FLAME, to provide geometric structure and controllable driving signals [4], [32], [33]. They estimate per-frame pose/expression coefficients through learning-based regressors or analysis-by-synthesis fitting [5]–[7], [34], [35], and then optimize a subject-specific renderer across frames. Such model-based formulations offer stable control and temporal coherence, but appearance recovery remains difficult under monocular ambiguity, especially for high-frequency textures and weakly constrained non-facial regions [30]. Related head/avatar creation works also explore complementary input modalities and priors, including semantic-constrained single-image head reconstruction, UV-UNet-based morphable head creation, and instruction-driven articulated avatar generation [36]–[38].

NeRF-style methods increase appearance capacity by modeling a continuous radiance field [8]. Representative monocular head-avatar systems, including NerFACE [9], IMAvatar [39], INSTA [29], Havatar [14], LatentAvatar [40], BakedAvatar [41], and Neural Head Avatars [42], achieve strong realism by conditioning radiance on pose and expression. However, volumetric ray marching remains computationally expensive for both training and inference. Several works reduce this cost with compact parameterizations such as multi-resolution hashing, tensor factorization, or motion-aware neural voxels [43]–[46], yet the sampling-and-compositing nature of volume rendering still makes high-resolution real-time reenactment challenging. These limitations motivate explicit rasterization-based representations that preserve controllability while improving efficiency.

B. 3D Gaussian Head Avatars

3D Gaussian Splatting (3DGS) replaces volumetric ray marching with efficient rasterization of anisotropic Gaussian primitives, offering an appealing quality–speed trade-off [17]. Recent head-avatar methods build on this backbone by coupling Gaussians with parametric motion, mesh binding, or canonical-space deformation. GaussianAvatars [18] binds Gaussians to a deformable template for controllable animation, FlashAvatar [19] parameterizes Gaussians in UV space for efficient monocular optimization, and FATE [25] extends monocular Gaussian avatars to full-head reconstruction and editing. SplattingAvatar [20] emphasizes mesh-embedded correspondences, PSAvatar [21] introduces a point-based shape model for expression-aware geometry, and related works explore blendshape-style Gaussian bases, progressive monocular training, or richer capture settings [22]–[24], [26], [27], [47].

These methods substantially improve efficiency, controllability, and geometric modeling, but a fidelity gap still remains in the monocular setting. In particular, many 3DGS head avatars continue to model appearance primarily as free per-Gaussian RGB values or low-order coefficients optimized directly from reconstruction losses. This design is simple and effective, but under monocular ambiguity it leaves point-wise appearance learning weakly constrained: different local explanations can produce similarly good renderings, encouraging over-smoothed solutions and weakened fine-scale cues. Our work focuses on this appearance bottleneck rather than on a new deformation or rendering mechanism. We introduce a frozen subject-specific prior distilled from a pretrained encoder (DINOv2 [28]) and use it to predict, rather than freely fit, Gaussian RGB appearance. The resulting formulation is complementary to existing 3DGS avatar pipelines: it preserves the standard Gaussian rendering path while explicitly regularizing the stage where monocular over-smoothing arises.

III. Method

A. Overview

Given a monocular video of a target subject, HiFiAvatar learns a person-specific animatable head avatar that supports expression- and pose-driven reenactment. As argued in Section 1, our target is not the renderer itself, but the under-constrained appearance-learning problem that remains when Gaussian colors are fit directly from monocular reconstruction losses. Our key idea is therefore to regularize per-Gaussian RGB prediction with a frozen *identity/texture prior dictionary*, so that each Gaussian predicts its appearance under the guidance of subject-specific visual prototypes rather than behaving as an unconstrained free color parameter.

The pipeline has three stages. **(i) Prior dictionary construction.** The training video is encoded by a frozen pretrained visual backbone to obtain token-level features, which are aggregated into an identity/texture prior dictionary. **(ii) Prior-guided Gaussian appearance.** Each Gaussian forms a learned query from its canonical anchor, retrieves a compact appearance cue from the prior dictionary, fuses it with a learnable base embedding, and predicts its RGB appearance directly. **(iii) Gaussian rendering.** The predicted prior-guided RGB appearance is rendered by the standard Gaussian splatting rasterizer. Figure 2 provides an overview of the pipeline.

B. Animatable Gaussian Avatar Representation

Following 3D Gaussian Splatting [17], we represent an avatar by a set of P anisotropic Gaussians with learnable attributes. To obtain stable expression- and pose-driven motion from monocular video, we adopt a mesh-guided binding to a tracked parametric head model (e.g., FLAME [4]). To absorb subject-specific geometric deviations that are not fully captured by the tracked template, we additionally optimize lightweight corrective parameters on the template geometry and deformation bases while keeping all Gaussians bound to the same template-driven motion.

a) Surface binding.: Let $\mathbf{V}^t \in \mathbb{R}^{K\times 3}$ denote the deformed mesh vertices at frame t, and let $\mathcal{F}$ be the triangle topology. We sample P anchor points on the canonical mesh and keep their binding fixed during training. Each Gaussian i is associated with a triangle index $\tau_i \in \mathcal{F}$ and barycentric weights $\boldsymbol{\beta}_i \in \mathbb{R}^3$ with $\sum_{j=1}^{3} \beta_{i,j} = 1$. The mesh anchor position at frame t is

$$\mathbf{p}_i^t = \sum_{j=1}^{3} \beta_{i,j}\, \mathbf{V}_{\tau_i,j}^t. \tag{1}$$

We also compute a local surface frame for τ_i at frame t, represented by a rotation $\mathbf{q}_i^t$ (as a quaternion) that aligns the canonical local frame to the current mesh orientation.

b) Gaussian attributes and deformation.: Each Gaussian i maintains learnable parameters: opacity o_i, log-scales $\mathbf{s}_i \in \mathbb{R}^3$, a canonical rotation quaternion $\mathbf{r}_i$, and a learnable base appearance embedding $\mathbf{b}_i \in \mathbb{R}^B$. For each frame t, we transport the Gaussian orientation by the mesh frame:

$$\mathbf{r}_i^t = \mathbf{q}_i^t \otimes \mathbf{r}_i, \tag{2}$$

and modulate the scale according to the local mesh scaling γ_i^t (computed from the deformed triangle) as $\exp(\mathbf{s}_i) \cdot \gamma_i^t$.

c) Local 3D offset.: Beyond the surface anchor $\mathbf{p}_i^t$, we allow each Gaussian to deviate from the mesh by a learnable local offset. Specifically, we learn an unconstrained vector $\tilde{\boldsymbol{\delta}}_i \in \mathbb{R}^3$ and bound it with

$$\boldsymbol{\delta}_i = \lambda \tanh(\tilde{\boldsymbol{\delta}}_i), \tag{3}$$

where λ is a global scale. The per-frame 3D center is then

$$\boldsymbol{\mu}_i^t = \mathbf{p}_i^t + \mathbf{R}(\mathbf{q}_i^t)\, \boldsymbol{\delta}_i, \tag{4}$$

where $\mathbf{R}(\mathbf{q}_i^t)$ applies the quaternion rotation. This bounded local-offset formulation prevents Gaussians from drifting excessively away from the tracked surface under noisy monocular supervision while still allowing a mild local relaxation for structures such as hair or mouth-adjacent regions.

C. Prior-Guided Gaussian Appearance

a) Identity/texture prior dictionary.: Let $\{I_t\}_{t=1}^{T}$ denote the training video frames. We construct the prior dictionary $\mathbf{M}$ using only frames from the training split and never access test frames, avoiding any evaluation leakage. A pretrained

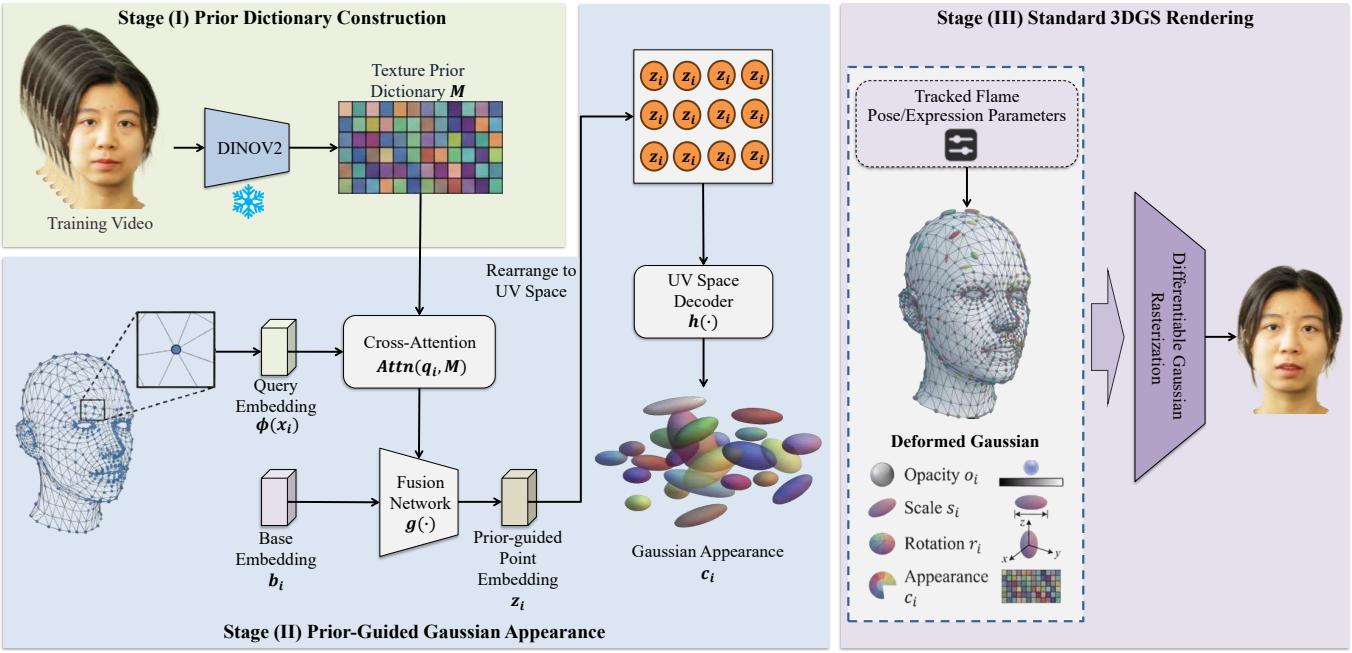

Fig. 2. Overview of HiFiAvatar. We construct a frozen identity/texture prior dictionary from the training video, let each Gaussian query this dictionary from canonical space, fuse the retrieved cue with a learnable base embedding, predict Gaussian RGB appearance, and render the final image with the standard 3DGS rasterizer.

vision encoder extracts token features from each frame. To obtain a stable subject-specific appearance prior and keep training efficient, we aggregate these tokens into a compact identity/texture prior dictionary

$$\mathbf{M} \in \mathbb{R}^{N \times D}, \tag{5}$$

where N is the number of dictionary entries and D is the token feature dimension. In practice, we use a frozen self-supervised vision transformer (DINOv2 [28]). Training frames are processed with a shared tokenization pipeline, including padding the image resolution to satisfy the ViT/14 input constraint, and we then average token features at the same grid index across training frames; the exact tokenization setup is given in the supplementary material. We do not interpret these averaged tokens as explicit 2D–3D correspondences. Instead, they serve as subject-specific appearance prototypes that remain fixed during avatar optimization. Because the dictionary is fixed throughout optimization, it cannot drift to absorb reconstruction noise; instead, it provides a stable subject-specific constraint on appearance prediction under monocular ambiguity.

b) Querying the prior dictionary.: Each Gaussian has a canonical 3D coordinate $\mathbf{x}_i$ defined in the avatar's canonical space (we use its canonical mesh anchor position, i.e., $\mathbf{p}_i^0$). A query embedding is computed from $\mathbf{x}_i$ by applying positional encoding followed by a two-layer MLP:

$$\mathbf{q}_i = \phi(\mathbf{x}_i). \tag{6}$$

A multi-head cross-attention module then retrieves an appearance cue from the prior dictionary:

$$\mathbf{a}_i = \mathrm{Attn}(\mathbf{q}_i, \mathbf{M}). \tag{7}$$

The attended feature $\mathbf{a}_i$ is first fused with the learnable base embedding $\mathbf{b}_i$ through a lightweight MLP:

$$\mathbf{z}_i = g([\mathbf{b}_i; \mathbf{a}_i]), \tag{8}$$

where $g(\cdot)$ produces a prior-guided point embedding. The learnable base embedding preserves per-point flexibility, while the retrieved prior cue prevents this flexibility from degenerating into unconstrained color fitting. Rather than regressing the RGB color of each Gaussian independently from $\mathbf{z}_i$, we further decode the fused point embeddings in canonical surface space. Specifically, we arrange $\{\mathbf{z}_i\}$ on the UV layout of the template and apply a lightweight convolutional decoder $h(\cdot)$ to predict Gaussian RGB appearance:

$$\{\mathbf{c}_i\} = h(\{\mathbf{z}_i\}), \qquad \mathbf{c}_i \in \mathbb{R}^3. \tag{9}$$

Concretely, this UV-space arrangement is defined once in canonical space by sampling a fixed set of template-bound anchors on the template UV map, where each Gaussian keeps the same associated template triangle and barycentric coordinates throughout training. The decoder therefore operates on a fixed canonical ordering of these sampled anchors rather than projecting deformed or offset 3D Gaussians back to the UV map at each frame. The learned local 3D offsets only affect the Gaussian centers used for rasterization after deformation, while the appearance embedding of each Gaussian remains attached to its original canonical template sample. This design avoids the ambiguity of dynamically mapping off-surface Gaussians into 2D and lets the UV-space CNN provide local appearance context over a stable template-aligned layout. This surface-space decoding allows neighboring Gaussians to exchange local appearance context before rasterization, which helps disambiguate local appearance under monocular supervision. In this way, the model combines a frozen subject-specific

Fig. 3. Qualitative comparisons on self-reenactment, cross-reenactment, and novel-view synthesis. Zoom-in crops highlight appearance fidelity in high-frequency regions.

prior with learnable per-point adaptation and local surface context, yielding a better-constrained appearance predictor under monocular supervision.

D. Appearance Prediction and Rendering

Given the current expression and pose, the Gaussian primitives are transformed to the deformed space and rendered by the standard 3DGS rasterizer. The prior-guided Gaussian colors are rasterized into the final image as

$$I = \mathcal{R}(\{\mathcal{G}_i\}, \{\mathbf{c}_i\}) \in \mathbb{R}^{3\times H\times W}, \tag{10}$$

where $\mathcal{R}$ denotes differentiable Gaussian rasterization and $H \times W$ is the target resolution. Our method keeps the standard 3DGS rasterizer unchanged and uses the prior only to constrain Gaussian RGB prediction before rasterization.

E. Training Objectives

We optimize all learnable parameters by minimizing an image reconstruction objective with regularization. For a training frame with ground truth image I^*, we render the prediction I

and minimize

$$\begin{aligned}\mathcal{L} = &\lambda_{\mathrm{rgb}}\|I - I^*\|_1 + \lambda_{\mathrm{vgg}}\sum_{\ell\in\mathcal{S}}\big\|\Phi_\ell(I) - \Phi_\ell(I^*)\big\|_1 \\ &+ \lambda_{\mathrm{dssim}}\big(1 - \mathrm{SSIM}(I, I^*)\big) + \lambda_{\mathrm{lpips}}\,\mathrm{LPIPS}(I, I^*) \\ &+ \lambda_{\mathrm{lap}}\mathcal{L}_{\mathrm{lap}} + \lambda_{\mathrm{scale}}\mathcal{L}_{\mathrm{scale}}.\end{aligned} \tag{11}$$

Here $\Phi_\ell(\cdot)$ denotes a fixed VGG feature extractor at layers $\mathcal{S}$. Importantly, we do not impose an extra feature-matching loss on the prior dictionary itself. The regularization effect of the prior is implicit: it comes from constraining the appearance predictor through a frozen subject-specific representation while still training the full avatar with standard image-space supervision.

a) Geometry regularization.: Since we optimize lightweight corrective geometry together with Gaussian attributes, we regularize the corrected mesh with a Laplacian smoothness prior:

$$\mathcal{L}_{\mathrm{lap}} = \frac{1}{K}\big\|\mathbf{L}\mathbf{V}^t - \mathbf{L}\bar{\mathbf{V}}^t\big\|_2^2, \tag{12}$$

where $\mathbf{V}^t$ is the corrected mesh at frame t, $\bar{\mathbf{V}}^t$ is the corresponding tracked template before applying corrective deformation, $\mathbf{L}$ is the mesh Laplacian operator, and K is the number of mesh vertices. This term keeps corrective geometry changes smooth and prevents unstable local distortions under monocular supervision.

b) Scale regularization.: We penalize excessive Gaussian anisotropy by

$$\mathcal{L}_{\mathrm{scale}} = \frac{1}{P}\sum_{i=1}^{P}\max\left(0, \frac{\max(\boldsymbol{\sigma}_i)}{\min(\boldsymbol{\sigma}_i)} - \tau\right), \tag{13}$$

where $\boldsymbol{\sigma}_i = \exp(\mathbf{s}_i) \in \mathbb{R}^3$ are the per-axis scales and τ is a threshold. This term prevents a small number of highly elongated Gaussians from compensating for poor appearance predictions with unstable geometry.

IV. Experiments

A. Datasets

Experiments target the monocular setting and evaluate the rendering fidelity of expression/pose-driven animatable head avatars. Following FATE [25], we evaluate on 20 monocular portrait sequences from four datasets: 10 subjects from INSTA Dataset [29], 3 subjects from PointAvatar Dataset [30], 3 subjects from NerFace Dataset [9], and 4 subjects from EmoTalk3D Dataset [31]. We follow the preprocessing pipeline of FATE to obtain per-frame camera parameters and facial motion coefficients (via MICA [7] for INSTA and a DECA-based pipeline [5] for the remaining datasets). All frames are resized to 512×512 and composited on a constant background using the available alpha mattes, and we remove regions below the neck via segmentation as in FATE.

B. Implementation Details

We initialize camera and facial motion parameters from the FATE-style preprocessing outputs and jointly refine them during training. Unless otherwise specified, the main model uses a 256×256 UV sampling layout, corresponding to 256^2 bound Gaussians. The appearance module predicts Gaussian RGB values from a frozen DINOv2 ViT-B/14 prior dictionary constructed by averaging token features over the training split. Each Gaussian queries this dictionary from canonical space with a 256-dimensional query using 8-head cross-attention, evaluated in 55K-query chunks for efficiency, and the retrieved cue is fused with a 64-dimensional learnable per-point base embedding. The fused point embeddings are decoded on the UV grid with a lightweight CNN point decoder to directly predict the 3-channel Gaussian RGB appearance, without any additional image-domain refiner. For geometric flexibility, we use bounded local 3D Gaussian offsets with scale $\lambda = 0.05$. All training, evaluation, and FPS measurements are conducted on a single NVIDIA RTX 3090 Ti GPU. Unless otherwise specified, we keep the same data preparation and evaluation resolution across all four datasets and do not introduce additional dataset-specific masking or post-processing beyond the shared FATE protocol described above. Additional details on token extraction, prior construction, network architecture, and optimization hyperparameters are provided in the supplementary material.

C. Quantitative Results

We follow the preprocessing and evaluation protocol of FATE [25] and report PSNR, SSIM, and LPIPS averaged over all test frames. We compare against six recent 3DGS-based baselines. These include FlashAvatar [19], SplattingAvatar [20], MonoGaussianAvatar [22], GaussianAvatars [18], HRAvatar [26], and FATE [25]. Table I reports subject-level results on INSTA [29], while Table II reports results on PointAvatar [30], NerFace [9], and EmoTalk3D [31]; the Avg. columns give dataset-level averages. To keep the comparison as direct as possible, we align all quantitative results in Tables I and II to the same evaluation setup summarized above, including FATE-style preprocessing, the same subject splits, 512×512 evaluation resolution, constant-background compositing with the available alpha mattes, neck-region removal, and frame-level PSNR/SSIM/LPIPS evaluation. For FlashAvatar, SplattingAvatar, MonoGaussianAvatar, GaussianAvatars, and FATE, we report the numbers published in FATE [25], where these methods were evaluated under a unified FATE-style protocol. HRAvatar is not fully reported in FATE. We therefore use the reported INSTA results from [26] and reproduce the PointAvatar, NerFace, and EmoTalk3D results under the same data preparation and evaluation settings used for our method.

Tables I and II show that HiFiAvatar achieves the best average PSNR, SSIM, and LPIPS on INSTA, PointAvatar, and NerFace. On EmoTalk3D, it attains the best average SSIM and LPIPS while remaining competitive in PSNR. The most consistent gains appear in LPIPS, indicating that the proposed prior-guided appearance model better preserves subject-specific visual details that are often over-smoothed

TABLE I
FULL COMPARISON OF QUANTITATIVE RESULTS WITH STATE-OF-THE-ART METHODS ON INSTA DATASET. AVG. DENOTES THE UNWEIGHTED MEAN ACROSS SUBJECTS.

Metric	Method	INSTA Dataset										
		bala	biden	justin	malte_1	marcel	nf_01	nf_03	obama	person4	wojtek_1	Avg.
PSNR↑	FlashAvatar	28.04	29.63	27.42	27.54	29.32	24.47	27.24	26.92	21.65	30.66	27.29
	SplattingAvatar	27.58	28.64	26.43	28.01	28.05	24.11	26.02	25.46	20.84	31.23	26.64
	MonoGaussianAvatar	28.95	29.97	27.28	27.93	28.84	24.58	27.22	27.39	21.98	29.90	27.40
	GaussianAvatars	27.99	28.52	26.70	27.53	29.10	23.80	26.00	25.12	20.58	31.27	26.66
	FATE	28.17	30.06	27.11	28.40	29.07	24.69	27.46	27.04	22.01	31.25	27.53
	HRAvatar	33.10	31.70	**33.29**	29.28	26.58	28.95	29.68	33.24	26.54	31.26	30.36
	Ours	**35.81**	**35.73**	32.64	**35.56**	**33.43**	**33.01**	**35.62**	**34.61**	**28.25**	**37.95**	**34.26**
SSIM↑	FlashAvatar	0.9170	0.9595	0.9580	0.9310	0.9328	0.9303	0.9227	0.9413	0.9009	0.9530	0.9347
	SplattingAvatar	0.9168	0.9501	0.9553	0.9352	0.9265	0.9201	0.9137	0.9341	0.8959	0.9563	0.9304
	MonoGaussianAvatar	0.9297	0.9599	0.9601	0.9340	0.9316	0.9320	0.9200	0.9482	0.9090	0.9480	0.9373
	GaussianAvatars	0.9397	0.9548	0.9609	0.9401	0.9394	0.9286	0.9257	0.9398	0.9032	0.9635	0.9396
	FATE	0.9267	0.9659	0.9606	0.9434	0.9371	0.9344	0.9253	0.9524	0.9097	0.9600	0.9415
	HRAvatar	0.9473	0.9635	0.9687	0.9429	0.9352	0.9398	0.9334	0.9647	0.9278	0.9590	0.9482
	Ours	**0.9791**	**0.9854**	**0.9839**	**0.9792**	**0.9547**	**0.9731**	**0.9756**	**0.9828**	**0.9468**	**0.9856**	**0.9746**
LPIPS↓	FlashAvatar	0.0484	0.0267	0.0410	0.0418	0.0757	0.0768	0.0601	0.0375	0.1403	0.0296	0.0578
	SplattingAvatar	0.1284	0.0694	0.0902	0.0838	0.1330	0.1385	0.1286	0.0820	0.1992	0.0659	0.1119
	MonoGaussianAvatar	0.0969	0.0547	0.0585	0.0758	0.1352	0.1146	0.0980	0.0532	0.1471	0.0534	0.0887
	GaussianAvatars	0.0618	0.0458	0.0616	0.0607	0.0943	0.1025	0.0827	0.0606	0.1626	0.0445	0.0777
	FATE	0.0534	0.0341	0.0445	0.0414	0.0760	0.0811	0.0674	0.0426	0.1298	0.0324	0.0603
	HRAvatar	0.0451	0.0306	0.0367	0.0476	0.0992	0.0868	0.0649	0.0279	0.0940	0.0358	0.0569
	Ours	**0.0149**	**0.0129**	**0.0213**	**0.0151**	**0.0544**	**0.0305**	**0.0208**	**0.0139**	**0.0596**	**0.0112**	**0.0255**

TABLE II
FULL COMPARISON OF QUANTITATIVE RESULTS WITH STATE-OF-THE-ART METHODS ON POINTAVATAR DATASET, NERFACE DATASET, AND EMOTALK3D DATASET. AVG. DENOTES THE UNWEIGHTED MEAN ACROSS SUBJECTS IN EACH DATASET.

Metric	Method	PointAvatar Dataset				NerFace Dataset				EmoTalk3D Dataset				
		yufeng	marcel	soubhik	Avg.	person1	person2	person3	Avg.	subject1	subject2	subject3	subject4	Avg.
PSNR↑	FlashAvatar	25.85	26.67	26.84	26.45	30.44	31.47	32.24	31.38	25.28	29.96	21.41	25.28	25.48
	SplattingAvatar	25.09	25.57	23.61	24.76	27.35	28.58	32.10	29.34	25.16	28.23	20.98	23.81	24.55
	MonoGaussianAvatar	28.50	27.01	28.98	28.16	32.09	33.89	35.35	33.78	25.31	30.22	21.46	24.74	25.43
	GaussianAvatars	25.12	25.15	23.27	24.51	27.03	28.62	31.54	29.06	25.12	27.71	20.77	23.10	24.18
	FATE	29.36	27.58	29.28	28.74	32.91	33.65	34.54	33.70	26.13	30.96	21.56	26.36	26.25
	HRAvatar	24.91	22.76	26.05	24.57	28.63	31.19	31.55	30.46	**29.65**	30.05	**24.22**	**28.38**	**28.08**
	Ours	**30.41**	**28.81**	**29.88**	**29.70**	**33.81**	**34.00**	**36.58**	**34.80**	26.57	**31.63**	21.22	26.87	26.57
SSIM↑	FlashAvatar	0.8863	0.9224	0.9221	0.9103	0.9620	0.9734	0.9569	0.9641	0.9218	0.9411	0.9087	0.9037	0.9188
	SplattingAvatar	0.8761	0.9056	0.8903	0.8907	0.9416	0.9516	0.9509	0.9480	0.9250	0.9350	0.9164	0.9020	0.9196
	MonoGaussianAvatar	0.9259	0.9374	0.9448	0.9360	0.9719	0.9812	0.9763	0.9765	0.9434	0.9453	0.9126	0.8827	0.9210
	GaussianAvatars	0.8938	0.9200	0.9095	0.9078	0.9504	0.9613	0.9560	0.9559	0.9394	0.9381	0.9200	0.9028	0.9251
	FATE	0.9239	0.9341	0.9418	0.9333	0.9716	0.9802	0.9691	0.9736	0.9429	0.9530	**0.9208**	0.9265	0.9358
	HRAvatar	0.8545	0.8827	0.8868	0.8746	0.9393	0.9542	0.9492	0.9476	0.9186	0.9181	0.8972	0.8915	0.9063
	Ours	**0.9368**	**0.9467**	**0.9528**	**0.9454**	**0.9777**	**0.9833**	**0.9796**	**0.9802**	**0.9530**	**0.9542**	0.9179	**0.9310**	**0.9390**
LPIPS↓	FlashAvatar	0.1043	0.1021	0.0607	0.0890	0.0377	0.0217	0.0317	0.0304	0.0668	0.0435	0.0827	**0.0787**	0.0679
	SplattingAvatar	0.1502	0.1510	0.1490	0.1501	0.0835	0.0660	0.0640	0.0712	0.1403	0.0877	0.1109	0.1484	0.1218
	MonoGaussianAvatar	0.0993	0.1280	0.0656	0.0976	0.0473	0.0224	0.0249	0.0315	0.0758	0.0546	0.0770	0.0921	0.0749
	GaussianAvatars	0.1287	0.1321	0.1163	0.1257	0.0681	0.0415	0.0432	0.0509	0.0806	0.0590	0.0912	0.1035	0.0836
	FATE	0.0694	0.0876	0.0586	0.0719	0.0329	0.0186	0.0256	0.0257	0.0705	0.0414	0.0843	0.0842	0.0701
	HRAvatar	0.1521	0.1881	0.1008	0.1470	0.0863	0.0500	0.0509	0.0624	0.1099	0.0720	0.1162	0.1331	0.1078
	Ours	**0.0519**	**0.0720**	**0.0331**	**0.0523**	**0.0243**	**0.0150**	**0.0157**	**0.0183**	**0.0350**	**0.0265**	**0.0734**	0.0793	**0.0536**

by monocular Gaussian avatar reconstruction. At the same time, the improvements in PSNR and SSIM show that these perceptual gains do not come at the expense of frame-level fidelity. Table III further shows that this improvement does not come with a prohibitive runtime cost. Because the prior-guided appearance branch predicts per-Gaussian RGB values from fixed canonical point samples, learned point embeddings, and the frozen DINO token dictionary, these RGB values are frame-independent for a trained subject. During inference, we therefore compute them once and cache them for the subject, which avoids repeated cross-attention and UV-space decoding during FPS measurement while still retaining the per-frame FLAME deformation, Gaussian transformation, and standard 3DGS rasterization costs. With this cached inference path, HiFiAvatar still runs at 230 FPS, preserving the practical efficiency of Gaussian avatar rendering while remaining much faster than MonoGaussianAvatar and SplattingAvatar. Overall, the cross-dataset and cross-metric results support our central claim: constraining Gaussian appearance prediction with a frozen subject-specific prior leads to a better-constrained and more detail-faithful appearance solution under monocular supervision.

D. Qualitative Results

Qualitative comparisons further support the quantitative gains. All qualitative baseline visualizations are reproduced by

TABLE III
INFERENCE SPEED COMPARISONS ON THE POINTAVATAR SUBJECT *soubhik*. ALL METHODS ARE MEASURED ON A SINGLE NVIDIA RTX 3090 TI GPU.

Method	FPS↑
FlashAvatar	262.91
SplattingAvatar	29.06
MonoGaussianAvatar	12.17
GaussianAvatars	289.69
HRAvatar	173.00
FATE	284.73
Ours	**230.00**

TABLE IV
ABLATIONS OF THE PRIOR-GUIDED APPEARANCE MODULE ON REPRESENTATIVE SUBJECTS FROM POINTAVATAR DATASET [30].

Variant	PointAvatar (*yufeng*)			PointAvatar (*soubhik*)		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Full (HiFiAvatar)	30.41	0.9368	0.0519	29.88	0.9528	0.0331
w/o prior	28.46	0.9004	0.0897	26.76	0.9304	0.0698
w/o query	29.13	0.9235	0.0659	27.95	0.9381	0.0523
w/o local context	29.80	0.9307	0.0615	28.88	0.9454	0.0456

us using their public implementations or checkpoints whenever available, with the same preprocessed inputs and rendering resolution for side-by-side comparison. Figure 3 presents three complementary evaluations: self-reenactment on held-out test frames, cross-reenactment driven by another subject, and novel-view synthesis of the reconstructed avatar. These settings probe reconstruction fidelity, motion-driven rendering quality under cross-reenactment, and view consistency, respectively. Across all three cases, HiFiAvatar preserves sharper and more stable appearance in the zoomed high-frequency regions than existing monocular Gaussian avatar baselines, which often exhibit over-smoothing or unstable texture in these areas. The visual comparisons are consistent with the quantitative results, suggesting that HiFiAvatar learns a more reliable appearance representation for high-frequency detail recovery under monocular supervision. Additional qualitative results are provided in the supplementary material.

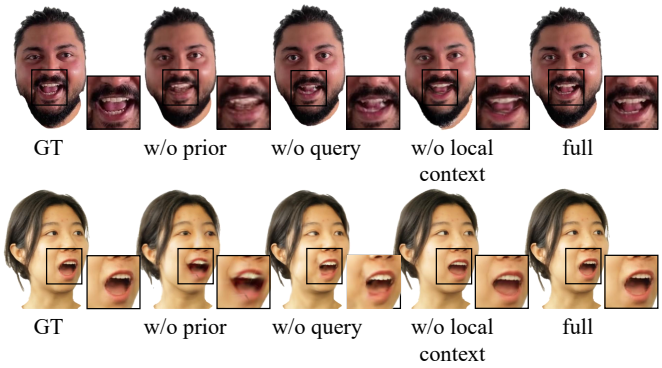

Fig. 4. Ablation visualization on *yufeng* and *soubhik*. We compare the full model with *w/o prior*, *w/o query*, and *w/o local context*.

E. Ablations

Ablation studies analyze the principal components of the prior-guided appearance formulation. We report ablations on two representative subjects from PointAvatar Dataset [30]: *yufeng* and *soubhik*.

a) Component ablations.: Table IV evaluates three questions: whether the frozen subject-specific prior is necessary, whether point-wise retrieval from this prior matters, and whether local surface-space context is beneficial. For fair comparison, all variants use the same initialization, the same number of Gaussians, and the same training schedule. *w/o prior* removes the dictionary branch entirely and falls back to appearance learning without frozen subject-specific prototypes. *w/o query* keeps the dictionary but replaces query-based retrieval with a shared mean token, weakening spatially varying prior guidance. *w/o local context* replaces the surface-space decoder with an independent point-wise decoder, removing neighborhood interaction before rasterization. Table IV shows that the full model performs best on both subjects across all three metrics. Removing the frozen prior causes the largest degradation: averaged over *yufeng* and *soubhik*, PSNR drops from 30.15 to 27.61, SSIM drops from 0.9448 to 0.9154, and LPIPS rises from 0.0425 to 0.0798. This confirms that the subject-specific prior dictionary is the primary source of regularization for monocular appearance learning. Replacing query-based retrieval with a shared mean token also leads to a clear performance drop (28.54 PSNR, 0.9308 SSIM, 0.0591 LPIPS on average), indicating that the prior should be selected in a point-dependent manner rather than injected globally. Replacing the surface-space decoder with a point-wise decoder produces a smaller but still consistent degradation (29.34 PSNR, 0.9381 SSIM, 0.0536 LPIPS on average), showing that UV-space neighborhood aggregation provides complementary local-context cues beyond the prior itself. Figure 4 qualitatively compares the full model against all three ablation variants.

V. LIMITATIONS

Our method still relies on sufficiently accurate camera and facial-motion tracking; noticeable errors in the tracked parameters can directly degrade Gaussian alignment and the quality of prior-guided appearance reconstruction. Moreover, the current monocular setting does not support true 360-degree full-head reconstruction, since heavily unobserved regions, especially the back of the head, remain weakly constrained by the input video. Finally, the reconstructed result is represented as an animatable Gaussian avatar rather than an explicit mesh, so the current system does not directly provide a mesh-based output for downstream applications that require surface geometry. Additional failure cases are provided in the supplementary material.

VI. CONCLUSION

This work presents HiFiAvatar, a framework for high-fidelity animatable head avatars from monocular video that targets a key appearance bottleneck in monocular 3DGS avatars. Our central idea is to replace unconstrained direct

Gaussian color fitting with prior-guided Gaussian RGB prediction regularized by a frozen identity/texture prior dictionary distilled from pretrained visual tokens. This design improves detail recovery and high-frequency appearance fidelity while preserving the standard 3D Gaussian splatting rendering pipeline, yielding the best average metrics on INSTA Dataset [29], PointAvatar Dataset [30], and NerFace Dataset [9], as well as the best average SSIM and LPIPS on EmoTalk3D Dataset [31].

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